Project 2 on Neural networks

Tim Herrmann Holmen

FYS-STK4155
Department of Physics

This project ...

CONTENTS

I. Introduction 2

II. Theory 2
A. Cost/ Loss functions and their gradients 2
1. MSE 2
2. Binary cross entropy: log loss 2
3. Multiclass cross entropy cost function 2
B. Activation functions 2
1. Sigmoid logit function 2
2. RELU function 2
3. leaky RELU function 2

III. Methods 2

IV. Results 2

V. Discussion 2

VI. Conclusion 2

I. INTRODUCTION

II. THEORY

A. Cost/ Loss functions and their gradients

1. MSE

with and without L_1 and L_2 norms, Regression problems.

2. Binary cross entropy: log loss

For binary classification problems with and without the L_1 and L_2 norms.

3. Multiclass cross entropy cost function

Softmaxx cross entropy or softmax loss function.

B. Activation functions

1. Sigmoid logit function

2. RELU function

3. leaky RELU function

III. METHODS

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION